

Olist E-Commerce Data Analysis — Business Report

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Project objective: analyze customers, orders, products, sales and delivery performance using the supplied Olist datasets and provide evidence-based business insights.

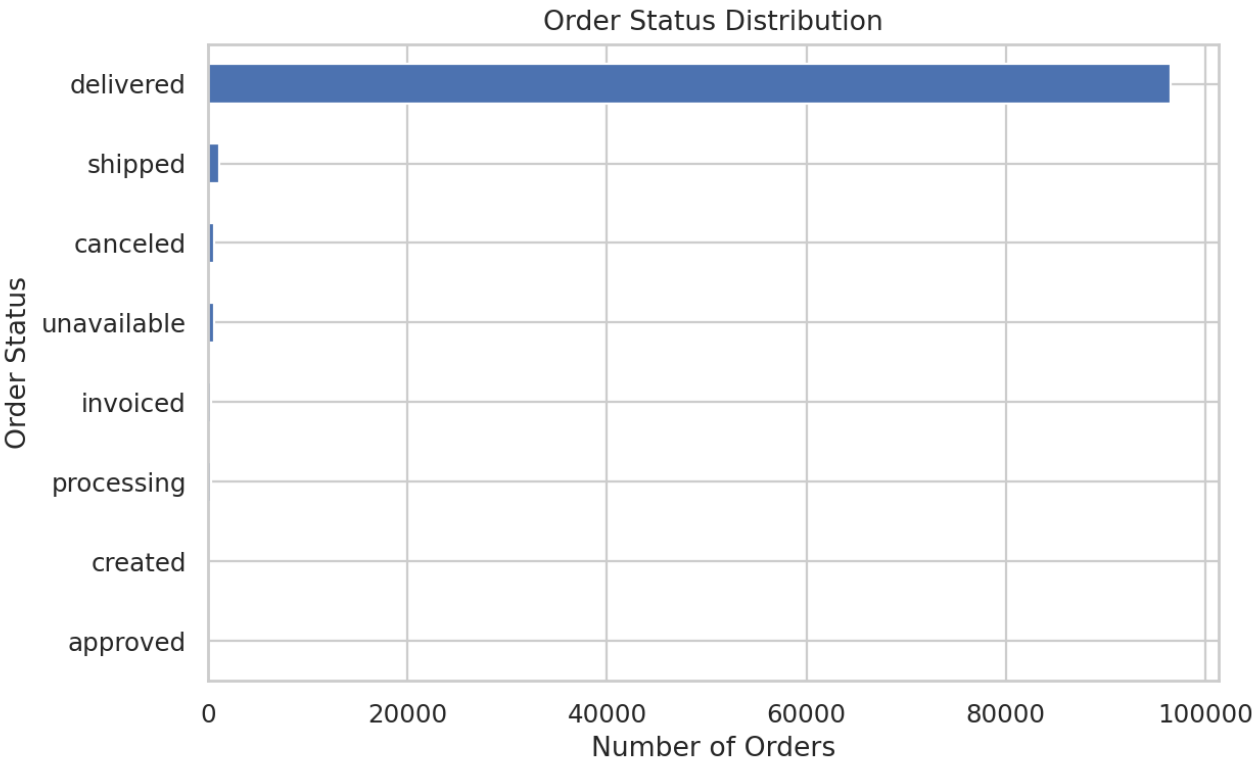
1. Data understanding and cleaning

The analysis uses 99,441 orders, 112,650 order-item records, 32,951 products and 96,096 unique customers. All four datasets contain zero exact duplicate rows. Order and shipping date fields were converted to datetime. Missing product categories were labeled unknown, and small numeric product-data gaps were median-imputed. Delivery dates were not artificially filled because missing delivery dates can represent orders that had not reached that stage.

2. Key numerical results

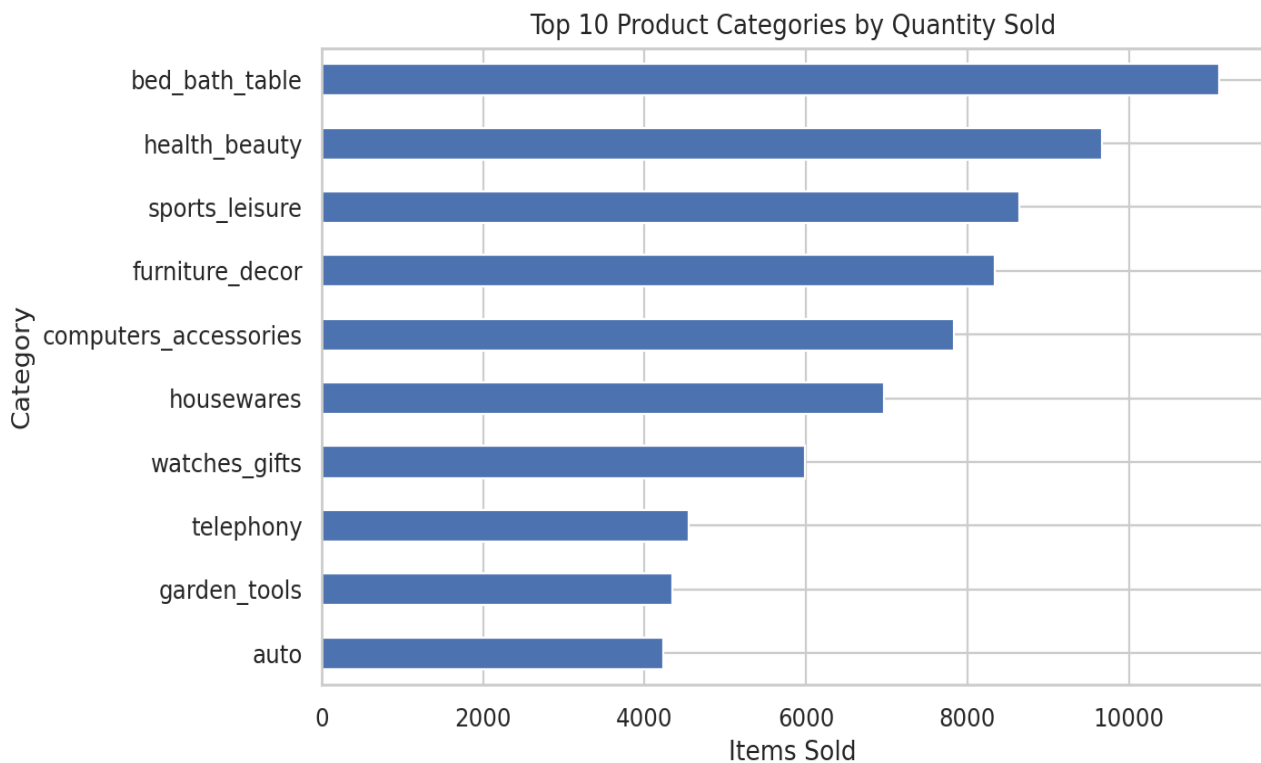
Metric	Result
Total orders	99,441
Unique customers	96,096
Unique products	32,951
Product sales	13,591,643.70
Freight value	2,251,909.54
Average product price	120.65
Average freight	19.99
Delivered orders	96,478
Average delivery time	12.56 days
Late delivered orders	7,826 (8.11%)

3. Order status

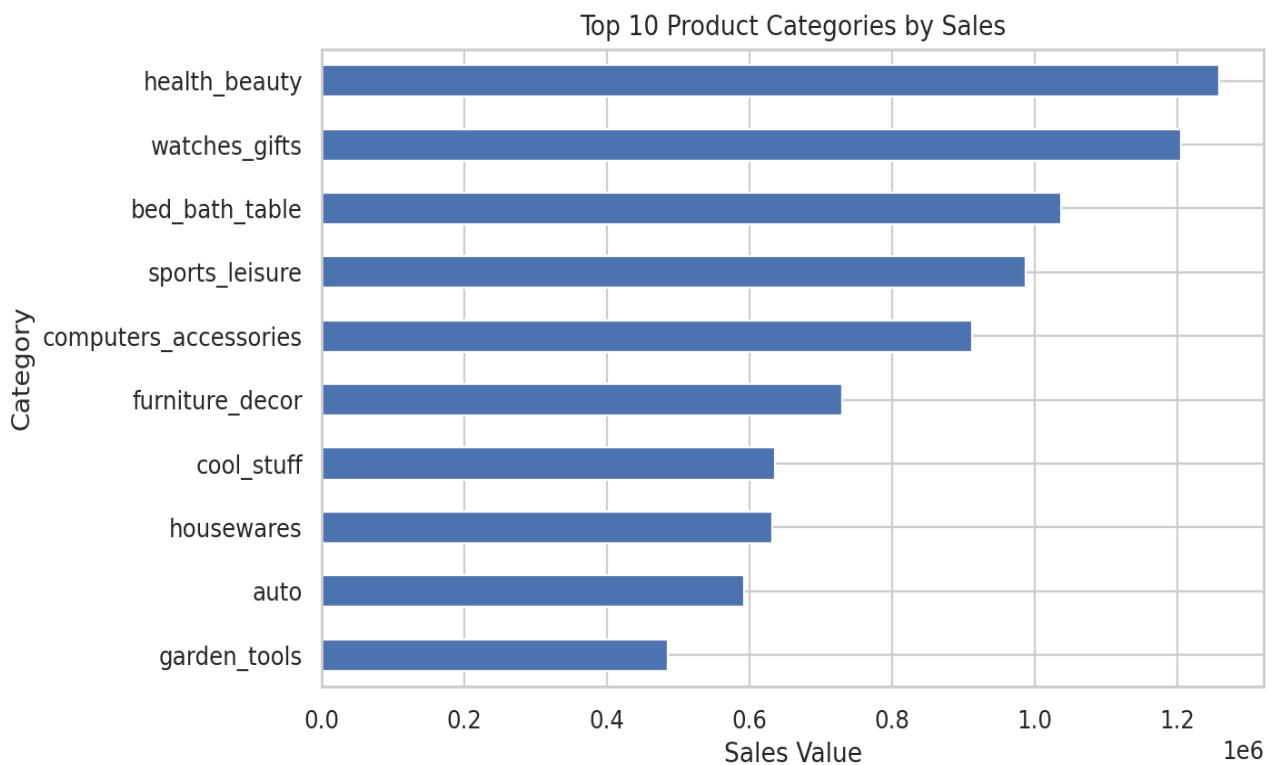


Delivered orders account for 97.02% of all orders. Shipped, canceled and unavailable are much smaller shares.

4. Sales and category analysis

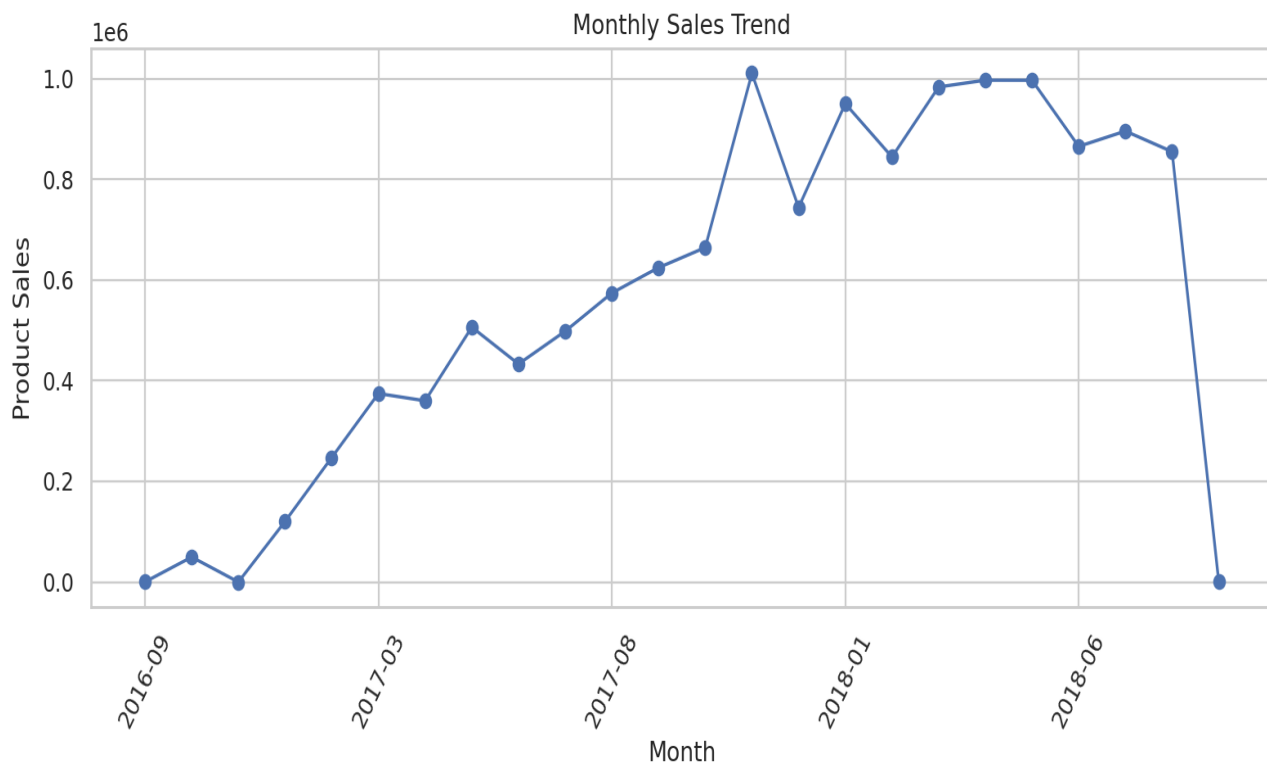


The leading quantity category is cama mesa banho (bed/bath/table), followed by health beauty and sports leisure.



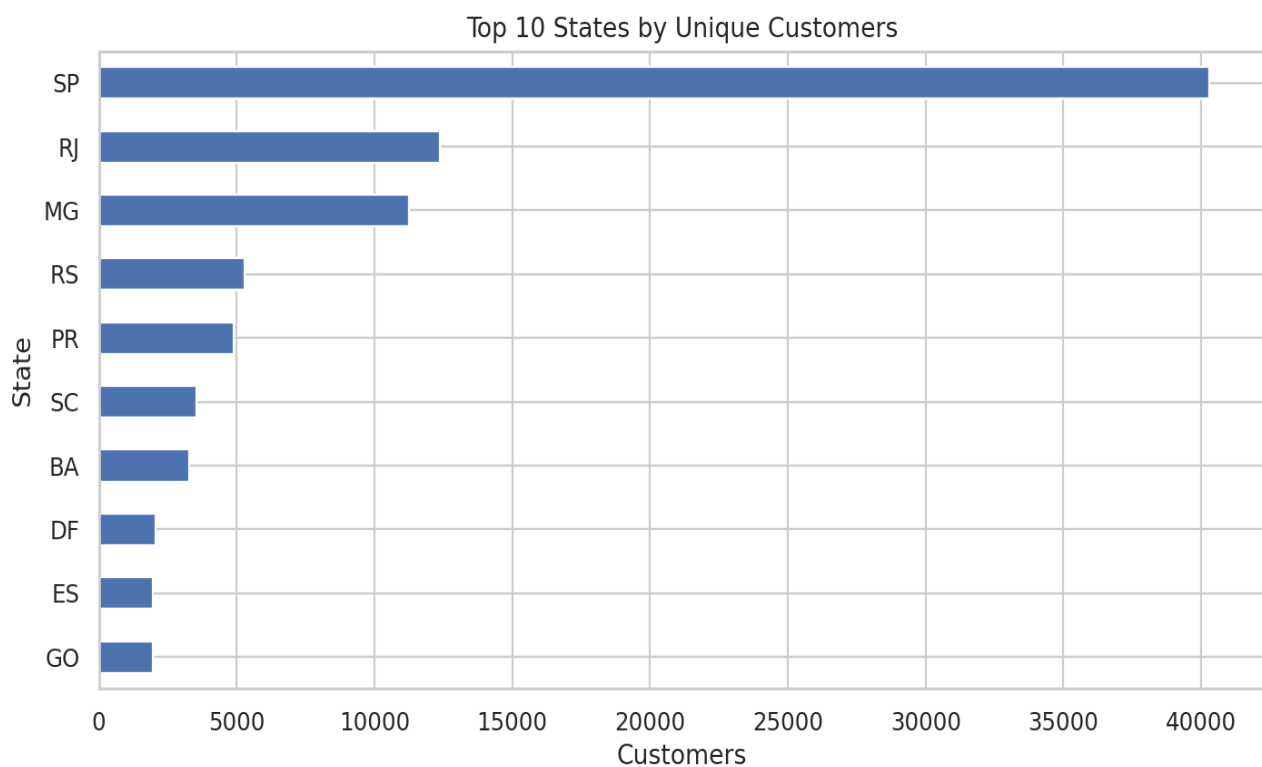
The highest-sales category is health_beauty, with 1,258,681.34 in product sales.

5. Monthly sales



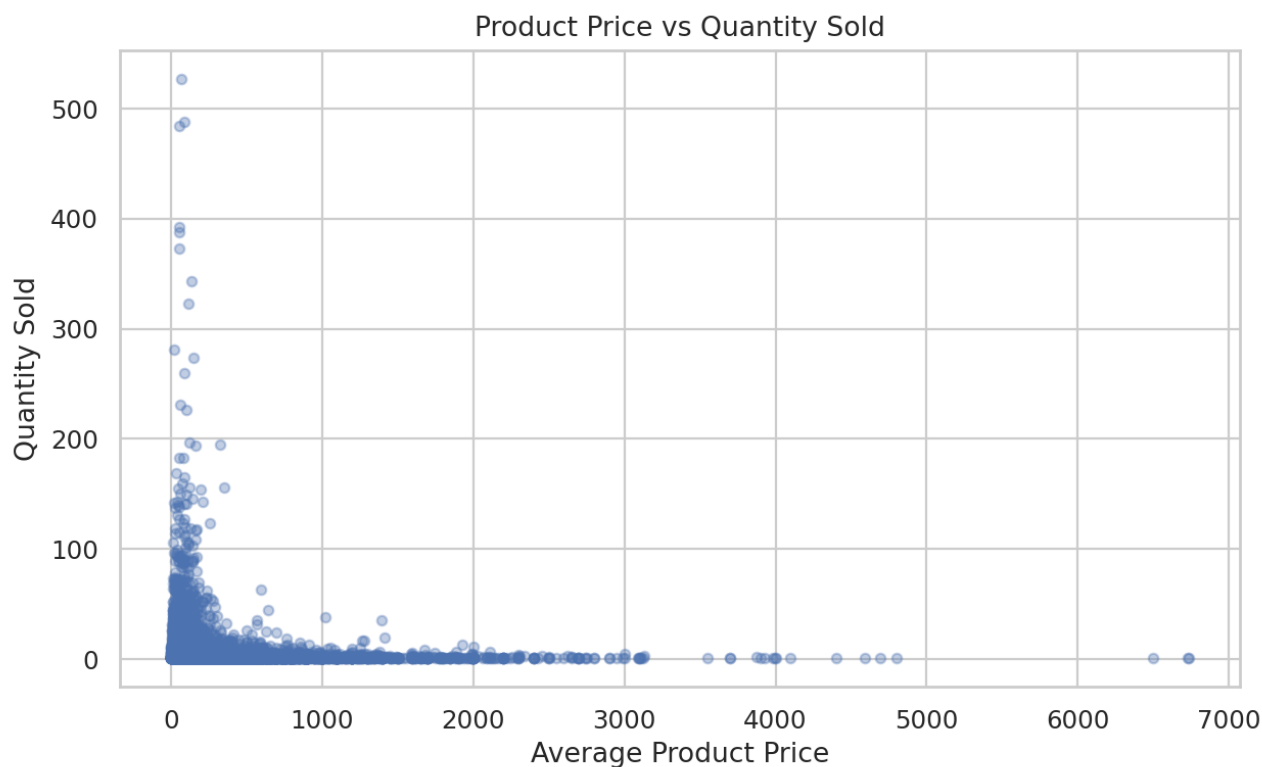
Highest month: 2017-11 (1,010,271.37). Lowest month: 2016-12 (10.90). The early and late months are partial periods and should be interpreted carefully.

6. Customer analysis



SP has 40,295 unique customers, representing 41.93% of all unique customers.

7. Price vs quantity



Product-level correlation between average price and quantity sold is -0.0321, indicating little linear association. This is an association measure, not evidence that price causes quantity changes.

8. Delivery performance



Among delivered orders with a recorded delivery date, mean delivery time is 12.56 days, median 10.22, minimum 0.53, and maximum 209.63. 7,826 (8.11%) were later than estimated.

9. Five business insights

Finding: Delivered orders represent 97.02% of all orders. Evidence: 96,478 delivered orders out of 99,441. Business implication: operational monitoring should pay attention to the smaller non-delivered statuses because they represent exceptions to the dominant flow.

Finding: health_beauty is the highest-sales category. Evidence: product sales of 1,258,681.34. Business implication: inventory and merchandising teams can monitor this category closely, while profitability should be checked separately.

Finding: Sales vary substantially by month. Evidence: November 2017 is the highest month at 1,010,271.37. Business implication: capacity and inventory planning can incorporate seasonal demand patterns.

Finding: SP contains the largest customer group. Evidence: 40,295 customers (41.93%). Business implication: regional service levels and campaign performance can be monitored closely in this state.

Finding: Delivery delays affect a measurable minority of delivered orders. Evidence: 7,826 orders (8.11%) arrived after the estimated date. Business implication: delivery exception analysis can focus on the late-order segment to identify operational patterns.

10. Five recommendations

1. Protect inventory for high-volume categories using the quantity ranking and stock monitoring.
2. Prioritize merchandising and marketing attention for high-sales categories, while adding margin data before making profitability decisions.
3. Use observed monthly peaks in capacity and inventory planning, especially around historically high-sales periods.
4. Investigate late deliveries by seller, carrier, destination state and order characteristics to identify repeat delay patterns.
5. Monitor customer concentration in high-volume states and test service or acquisition initiatives in lower-volume regions.